

Nishchay Deepak Mittal

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Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
B.Tech in Artificial Intelligence

2024 – Present
CGPA: 9.42/10

Academic Projects

SemEval-2026: Narrative Story Similarity | [GitHub](#)

2025-2026

- Engineered a **narrative similarity system** for SemEval-2026, achieving **62% accuracy** on the CodaBench leaderboard using **DeBERTa-v3**, **BGE-Large**, and **E5-v2** encoders.
- Expanded a **2,736-triplet** training dataset using **Label-Aware Augmentation** and **5-fold cross-validation**, improving validation accuracy to **79.16%** for the **DeBERTa-v2 Siamese** variant.
- Developed a **multi-objective loss function** (**Margin Ranking + Contrastive**) with a 0.35 margin and 0.07 temperature scale, generating **1024-dimensional L2-normalized embeddings**.
- Developed an ensemble of **5+ neighbor models** (**BGE-Large/Base**, **E5**, **SBERT**, **TF-IDF**), utilizing **arithmetic mean aggregation** to yield a **+0.5% gain** in retrieval stability.
- Accelerated training of **335M-parameter models** on **NVIDIA H100/T4 GPUs** via **Mixed Precision (AMP)** and **Layer Freezing** (top 4 layers) to manage **15GB VRAM** constraints.

Gen-AI Content Creation | [Website](#) | [GitHub](#)

Team Project

- Designed a full-stack AI-powered content generation platform using Next.js and Django, collaborating in a **3-member team** to deliver **scalable and responsive features**.
- Integrated LangChain with LLMs (Groq) to generate **4+ content formats** including blogs, tweets, YouTube scripts, and social media captions.
- Engineered a rich content workflow with CKEditor and PDF export, enabling users to **create, edit, and download content**.
- Implemented API integrations with Unsplash and Pexels to provide **high-resolution dynamic images**, enhancing overall user experience.

Mindsettler | [Website](#) | [GitHub](#)

Team Project

- Built a full-stack **mental wellness platform** using **Next.js (React + TypeScript)** and **Django**, delivering scalable **REST APIs** and a responsive user interface.
- Engineered a **custom NLP chatbot** using **PyTorch** and **sentence-transformers (all-MiniLM-L6-v2)**, leveraging **vector embeddings** and **cosine similarity** for high-accuracy intent classification.
- Optimized chatbot performance and latency by implementing **thread-safe Singleton caching** for models and embeddings in memory, enabling fast responses for concurrent users.

VeloCity – AI-Driven Parking Intelligence | [Website](#) | [GitHub](#)

Team Project

- Built a **traffic enforcement command center** using **FastAPI**, **Vanilla JS**, and **Tailwind CSS**, converting reactive patrols into **proactive, data-driven enforcement**.
- Applied **DBSCAN spatial clustering (Scikit-learn)** on violation data to detect illegal-parking hotspots and quantify their impact via a proprietary **Congestion Risk Score (CRS)**.
- Trained a **LightGBM** model to forecast next-day high-risk intersections, and integrated **Groq's Llama-3-8b** to auto-generate localized daily patrol briefs for field officers.
- Developed an **interactive 3D hotspot dashboard** with **Deck.gl**, **MapLibre GL JS**, and **Three.js**, featuring **voice-command search** via the Web Speech API and optimized patrol-route plotting.

Technical Skills

Languages: C, C++, Python

Web & Backend: Next.js, React.js, Django, FastAPI, HTML, CSS, Tailwind CSS

AI/ML: LangChain, HuggingFace, SBERT, DeBERTa, LightGBM, Scikit-learn

Libraries & APIs: Unsplash, Pexels, iNaturalist, Wikipedia API, Deck.gl, Three.js

Tools: Git, GitHub, VS Code, CKEditor

Achievements and Responsibilities

- Co-authored a research paper accepted at ACL 2026 (SemEval) on narrative story similarity detection using transformer models.
- Winner - WebWonders Hackathon, NEXUS SVNIT.
- Finalist - Genesis Hackathon organised by ACM, CEV, GDGC
- Representative - NEXUS, SVNIT Surat
- Executive - ACM, SVNIT Surat